「Def. Let X be a Metric Space, Y be a Normed Space.
A sequence of functions $\{f_n : X \to Y\}$ is called pointwise bounded on X if:

for any $x \in X$, $\{f_n(x)\}$ is bounded.

And it is called uniformly bounded if: for some $M > 0$,

$$\sup_X |f_n| < M \quad \text{for all } n \in \mathbb{N}.$$

「Thm. If $\{f_n\}$ is a pointwise bounded sequence of complex-valued functions on a countable set E, then $\{f_n\}$ has a pointwise convergent subsequence.
Proof. Denote $E = \{x_i \mid i \in \mathbb{N}\}$. Since $\{f_n(x_1)\}$ is bounded, it has a convergent subsequence, denote by $\{f_{1,k}\}$. Similarly, $\{f_{1,k}(x_2)\}$ has a convergent subsequence, denote by $\{f_{2,k}\}$. Inductively, we obtain

S_1 :	$f_{1,1}$	$f_{1,2}$	$f_{1,3}$	...	Converges at	x_1
S_2 :	$f_{2,1}$	$f_{2,2}$	$f_{2,3}$	... ↓ sub	//	x_1, x_2
S_3 :	$f_{3,1}$	$f_{3,2}$	$f_{3,3}$	.. ↓ sub	//	x_1, x_2, x_3
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And, the order of each functions is preserved regardless of they are contained every column, by the definition of Subsequence. Then, the sub sequence

$$S = \{f_{n,n}\}$$

Converges for each x_i. □
Note: $\mathcal{C}_b(X) = \{f : X \to \mathbb{C} \mid f \text{ continuous, bounded}\}$ where X is Metric Space.

「Lemma. If K is Compact Metric space, $f_n \in \mathcal{C}_b(X)$, $n \in \mathbb{N}$, and $f_n \rightrightarrows f$ on K, then $\{f_n\}$ is equicontinuous on K.
Proof. Using 3ε-Method. $f_n \rightrightarrows f \to$ Conti on Compact.

Arzelà-Ascoli Theorem.

Let X be a Compact Metric Space.

If $\{f_n\}$ is pointwise bounded and equicontinuous on X,

then $\{f_n\}$ is uniformly bounded and $\{f_n\}$ contains a uniformly convergent subsequence.

Proof. Given $\varepsilon>0$, there exists $\delta>0$ s.t. $d(x,y)<\delta \Rightarrow |f_n(x)-f_n(y)|<\varepsilon$ for all $n\in\mathbb{N}$.

Since X is compact, there are finitely many points $p_1,\cdots,p_r\in X$ s.t. $X=\bigcup_{i=1}^{r} B_\delta(p_i)$.

Since $\{f_n\}$ is pointwise bounded, there exist $M_i>0$ s.t. for all $n\in\mathbb{N}$, $|f_n(p_i)|<M_i$.

Now, for any $x\in X$, and $n\in\mathbb{N}$, $|f_n(x)| \le |f_n(x)-f_n(p_i)| + |f_n(p_i)| \le \varepsilon + \max\{M_i \mid i=1,\cdots,r\}$.

(for some p_i s.t. $x\in B_\delta(p_i)$)

To show that the second assertion, Compact + Metric $\Rightarrow$ 2nd-Countable $\hookrightarrow$ Separable.

Let $E\subseteq X$ be a Countable dense set. By the above Thm,

$\{f_n\}$ has a pointwise convergent subsequence on E, denote this by $\{f_{n_i}\}$. Simply, $g_i = f_{n_i}$.

Given $\varepsilon>0$, take $\delta>0$ s.t. $d(x,y)<\delta \Rightarrow |f_n(x)-f_n(y)|<\varepsilon$ for all $n\in\mathbb{N}$.

Since X is compact and E is dense in K, the set

$$\{B_\delta(x) \mid x\in E\} \text{ covers } X, \text{ and}$$

has a finite subcover, $\{B_\delta(x_i) \mid i=1,\cdots,m\}$. Since $\{g_i(x)\}$ converges for all $x\in X$,

there exists $N\in\mathbb{N}$ s.t. $i,j\ge N$, $1\le s\le m \Rightarrow |g_i(x_s)-g_j(x_s)|<\varepsilon$.

And, for any $x\in X$, for some $1\le s\le m$, $x\in B_\delta(x_s)$, thus $|g_i(x)-g_i(x_s)|<\varepsilon$ for all $i\in\mathbb{N}$.

Now, for all $x\in X$ and $i,j\ge N$,

$$|g_i(x)-g_j(x)| \le |g_i(x)-g_i(x_s)| + |g_i(x_s)-g_j(x_s)| + |g_j(x_s)-g_j(x)| < 3\varepsilon.$$

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